

Simarpreet Singh

Student Developer

Portfolio:

<https://www.simarpreet Singh.org>

GitHub:

<https://github.com/Sim20004>

Email: me@simarpreet Singh.org

Education

Queen Mary's
Grammar School,
Walsall
Year 9 Student

Personal Profile

Student developer with experience in Python, Rust, web development, Linux administration, and self-hosting. Creator of several software projects including a privacy-focused chat platform, a programming language, and an AI-assisted music learning system. Passionate about open source software and learning new technologies.

Technical Skills

Programming

- Python
- Rust
- HTML
- CSS
- Java

Technologies

- Linux
- Git
- GitHub
- Docker
- Nginx
- Cloudflare

In practice: Rust builds the Teapot compiler; Python and Linux run the self-hosted infrastructure behind Vlyra AI; HTML/CSS shape the portfolio site; Git, GitHub, Docker, Nginx, and Cloudflare handle version control and deployment across all of the projects below.

Qualifications & Achievements

Computer Science

OCR Entry Level
Certificate in Computer
Science (ELCS)
30/07/2026

Music

ABRSM Grades 1-4 Piano
(Practical) with
Distinction, Merit,
Distinction, and Merit
respectively
2023-2026
Trinity College Grade 4
Clarinet (Performance)
with Merit
12/02/2026

Mandarin

National Achievement
Hanyu Qiao Zhongwen
Shuo Bisai (China Bridge
National Mandarin
Speaking Competition)
organised
by Hanyu Qiao
Organisation
(China Bridge)
**First place in Primary
Category nationwide.**
26/10/2025

Projects - repositories are linked in portfolio above.

FOSSil Chat *(Ongoing)*

- Privacy-focused chat platform

Vlyra AI *(Ongoing)*

- Developing an AI-assisted music learning platform

Teapot Programming Language and TeapotOS *(Ongoing)*

- For Teapot Programming Language: Designed and implemented a custom programming language with lexical analyser, parser, and abstract syntax tree
- For TeapotOS: Wrote an operating system from scratch in Assembly, aiming to support Windows, Linux, and MacOS binaries

Personal Portfolio Site *(Ongoing)*

- Designed and developed personal website
- Implemented responsive design and SEO optimisation

Technical Experience

- Self-hosted Linux servers
- Docker container deployment and management
- Nginx web server configuration
- Cloudflare DNS and domain management
- Git and GitHub workflows

Leadership and Public Speaking

- Invited to deliver a speech to officials and representatives involved in UK-China relations
- Presented speech in both Mandarin and English

Languages

Punjabi

Native speaker

English

Native speaker

Mandarin

Intermediate proficiency
Participant in the Mandarin Excellence Programme (MEP)

Interests

- Software Development
- Open Source Software
- Linux & Self-Hosting
- Programming Language/Compiler Design
- Piano, Clarinet, and Saxophone
- Cybersecurity

References

Available upon request